



Instruction Manual for Electrical Technology Laboratory **Department of Electrical Engineering, IIT (ISM) Dhanbad**

Experiment No: 02

VERIFICATION OF THEVENIN'S AND NORTON'S THEOREMS



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Experiment No.**TITLE: Verification of Thevenin's and Norton's Theorems****OBJECTIVE:** To verify the Thevenin's and Norton's theorems experimentally.**APPARATUS USED:**

Serial No.	Name of the Equipment	Rating	Model No. of the Equipment	Makers Name
1.				
2.				
3.				
4.				

THEORY:**A. Thevenin's Theorem**

Statement:- Any two terminal, linear, bilateral network having a number of voltage, current sources and resistances can be replaced by a simple equivalent circuit consisting of a single voltage source in series with a resistance, where the value of the voltage source is equal to the open circuit voltage i.e Thevenin's voltage, V_{Th} across the two terminals of the network, and the resistance is the equivalent resistance i.e. Thevenin's resistance, R_{Th} measured between the terminals with all energy sources replaced by their internal resistances.

The current I_L through load resistance R_L can be calculated as, $I_L = \frac{V_{Th}}{R_{Th} + R_L}$

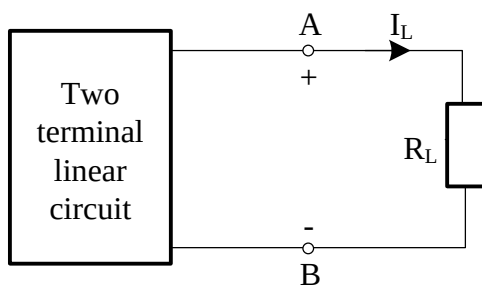


Fig. 1. Circuit diagram of a two terminal linear circuit.

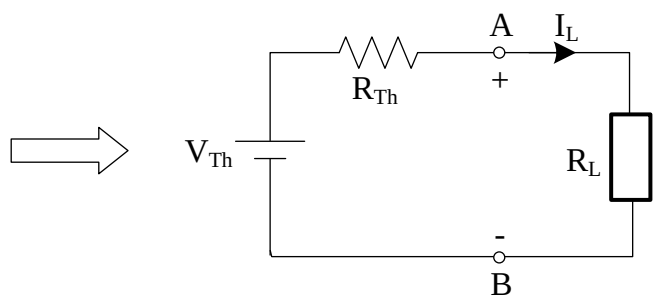


Fig. 2. Thevenin's equivalent circuit diagram.

B. Norton's Theorem

Statement:- Any two terminal, linear, bilateral network with current sources, voltage sources and resistances can be replaced by an equivalent circuit consisting of a current source in parallel with a resistance. The value

of the current source is the short circuit current i.e. Norton's current, I_N between the two terminals of the network and the resistance is the equivalent resistance i.e. Norton's resistance, R_N measured between the terminals of the network with all the energy sources replaced by their internal resistances.

The current I_L through load resistance R_L can be calculated as, $I_L = I_N \cdot \frac{R_N}{R_N + R_L}$

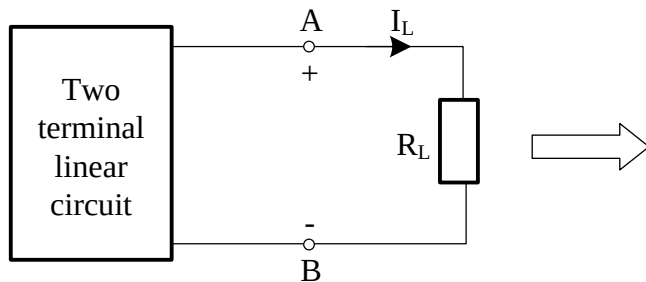


Fig. 3. Circuit diagram of a two terminal linear circuit.

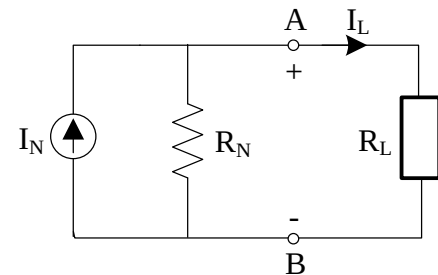


Fig. 4. Norton's equivalent circuit diagram.

CIRCUIT DIAGRAM:

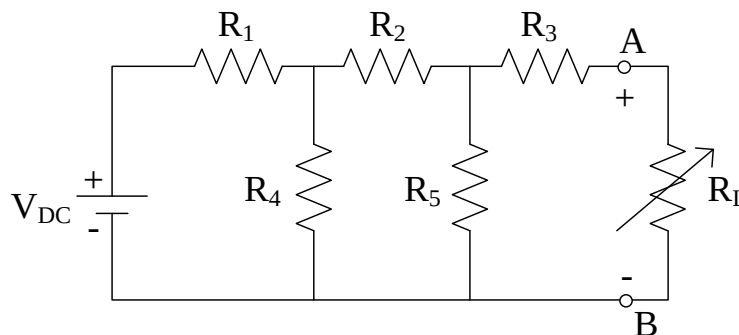


Fig. 5. Circuit diagram of the experimental setup.

PROCEDURE:

A. For verification of Thevenin's Theorem

- 1) Construct the circuit as shown in Fig. 5.
- 2) Switch the power supply and measure the open circuit voltage, V_{Th} across terminal A-B.
- 3) Short circuit the V_{DC} and calculate the equivalent resistance, R_{Th} across terminal A-B.
- 4) Construct Thevenin's equivalent circuit as shown in Fig. 2.
- 5) Calculate the load current, $I_L = \frac{V_{Th}}{R_{Th} + R_L}$ by varying the load resistance R_L .
- 6) Measure the load current, I_L' from the experimental setup.
- 7) Compare I_L and I_L' .
- 8) Calculate the experimental error.

B. For verification of Norton's Theorem

- 1) Construct the circuit as shown in Fig. 5.
- 2) Switch the power supply and measure the short circuit current, I_N .
- 3) Short circuit the V_{DC} and calculate the equivalent resistance, R_{Th} across terminal A-B.
- 4) Construct Norton's equivalent circuit as shown in Fig. 4.
- 5) Calculate the load current, $I_L = I_N \cdot \frac{R_N}{R_N + R_L}$ by varying the load resistance R_L .
- 6) Measure the load current, I_L' from the experimental setup.
- 7) Compare I_L and I_L' .
- 8) Calculate the experimental error.

COMMENTS/ DISCUSSIONS:

Write your comments on the results obtained and discuss the discrepancies, if any.

PRECAUTIONS:

- 1) Turn off the power switch to equipment before making connections.
- 2) Don't use broken connecting wires.
- 3) Maintain a work space clear of extraneous material such as books, papers, and clothes.
- 4) Don't wear loose clothes.
- 5) Wear shoes that cover the feet.

REPORT:

- 1) State the difference between linear and non-linear circuits. Give some examples of linear and non-linear elements in circuit theory.
- 2) State the difference between unilateral and bilateral circuits. Give some examples of unilateral and bilateral elements in circuit theory.
- 3) Draw the Thevenin's and Norton's equivalent circuit and write down the formula of the load current.
- 4) What are the limitations of Thevenin's and Norton's theorem?
- 5) Can we apply Thevenin's theorem in circuit having multi-source?

DATA SHEET

Experiment No.

Date:

TITLE: Verification of Thevenin's and Norton's Theorems

Name: _____ Sem: _____ Year: _____

Adm. No. _____

Experimental Data:-

Thevenin's Theorem:-

Sl. No.	Load Resistance, R_L (Ω)	Network under test	Open circuit the load terminal		Thevenin's equivalent circuit	Remarks
		Load current, I_L' (mA)	Thevenin's voltage, V_{Th} (V)	Thevenin's resistance, R_{Th} (Ω)	Load current, I_L (mA)	Experimental error (%)
1.						
2.						
3.						

Norton's Theorem:-

Serial No.	Load Resistance, R_L (Ω)	Network under test	Short circuit the load terminal		Norton's equivalent circuit	Remarks
		Load current, I_L' (mA)	Norton's current, I_N (mA)	Norton's resistance, R_N (Ω)	Load current, I_L (mA)	Experimental error (%)
1.						
2.						
3.						

 (Signature of the teacher)